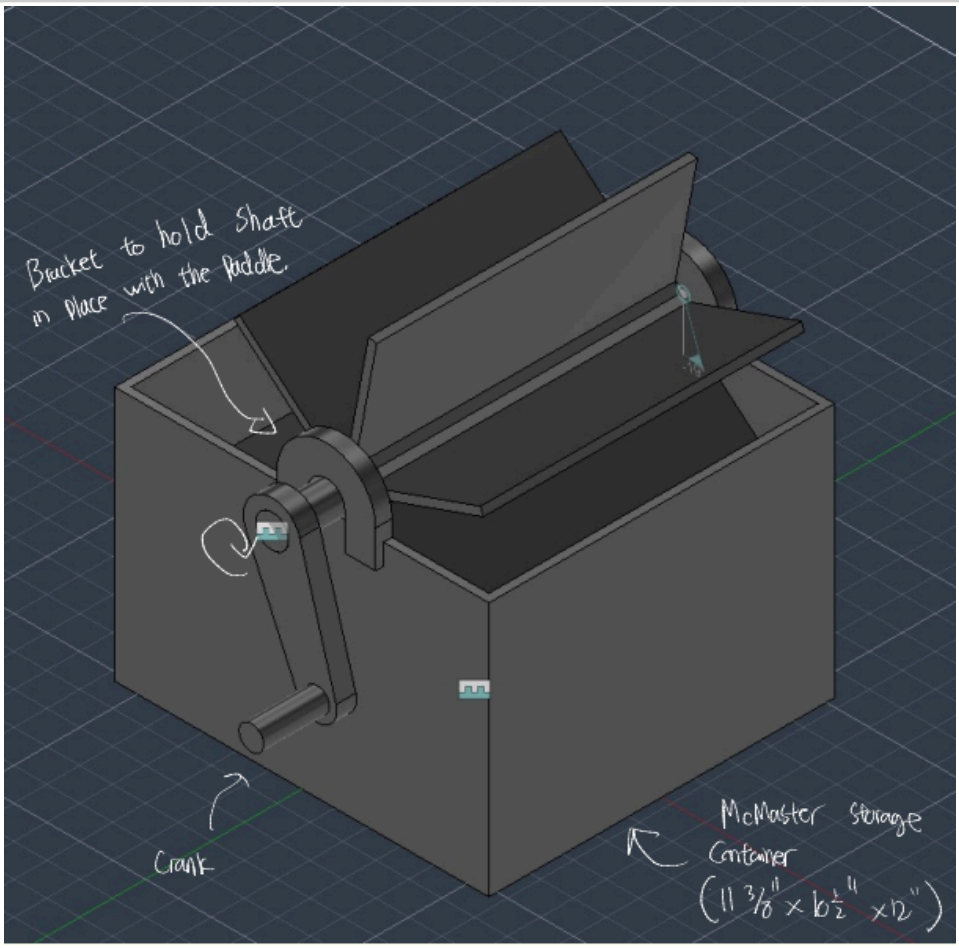
ODP 5: Functional Prototype; Team GrapeSoup

Design Documentation:

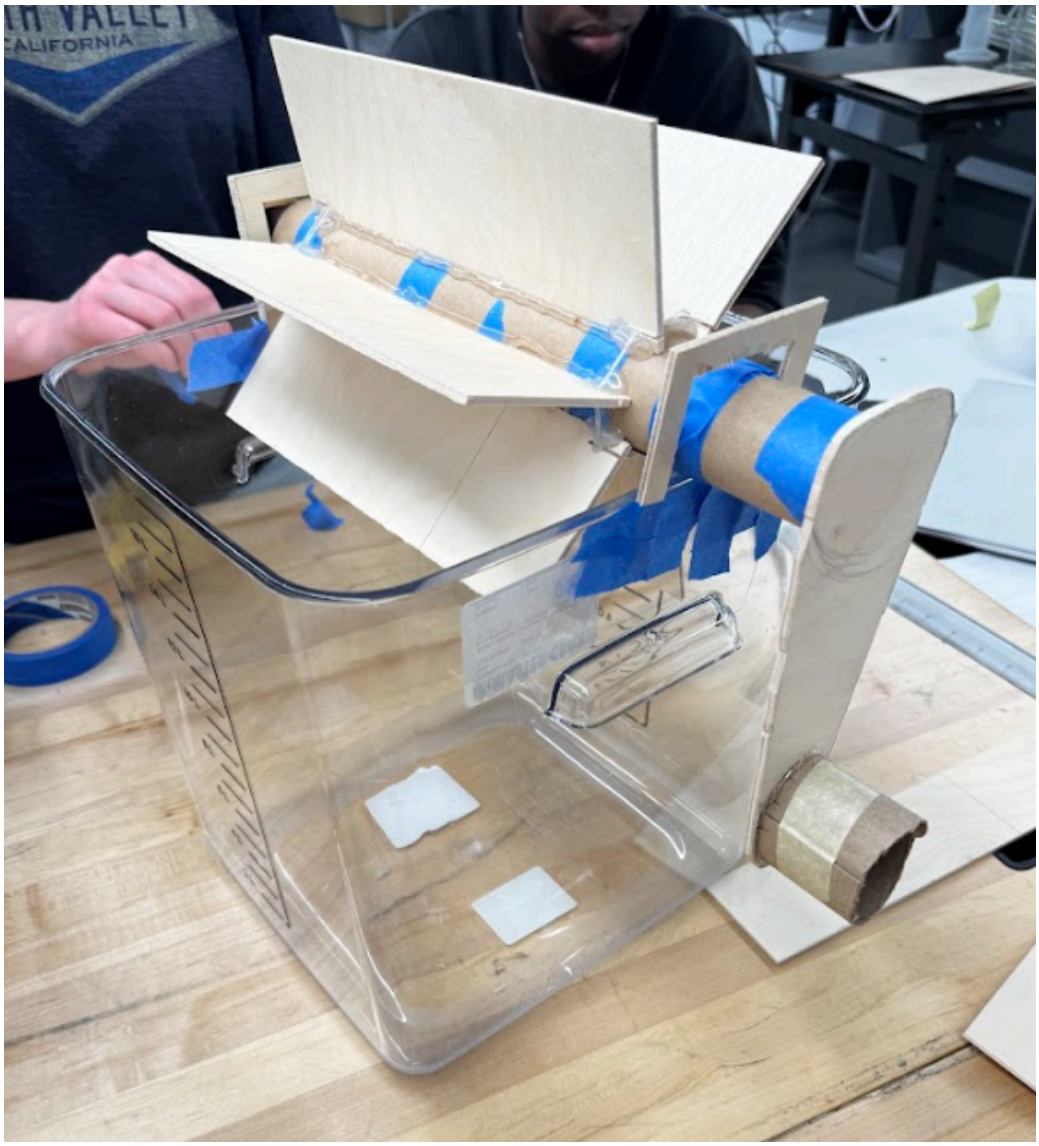
Description	McMaster Code	Quantity	Unit of measurement	Total Cost
Plastic storage containers for Food and Beverage, 4.5 gal, 11 3/8" x 10 1/2" x 12"	8993T46	1	Each	$45.23

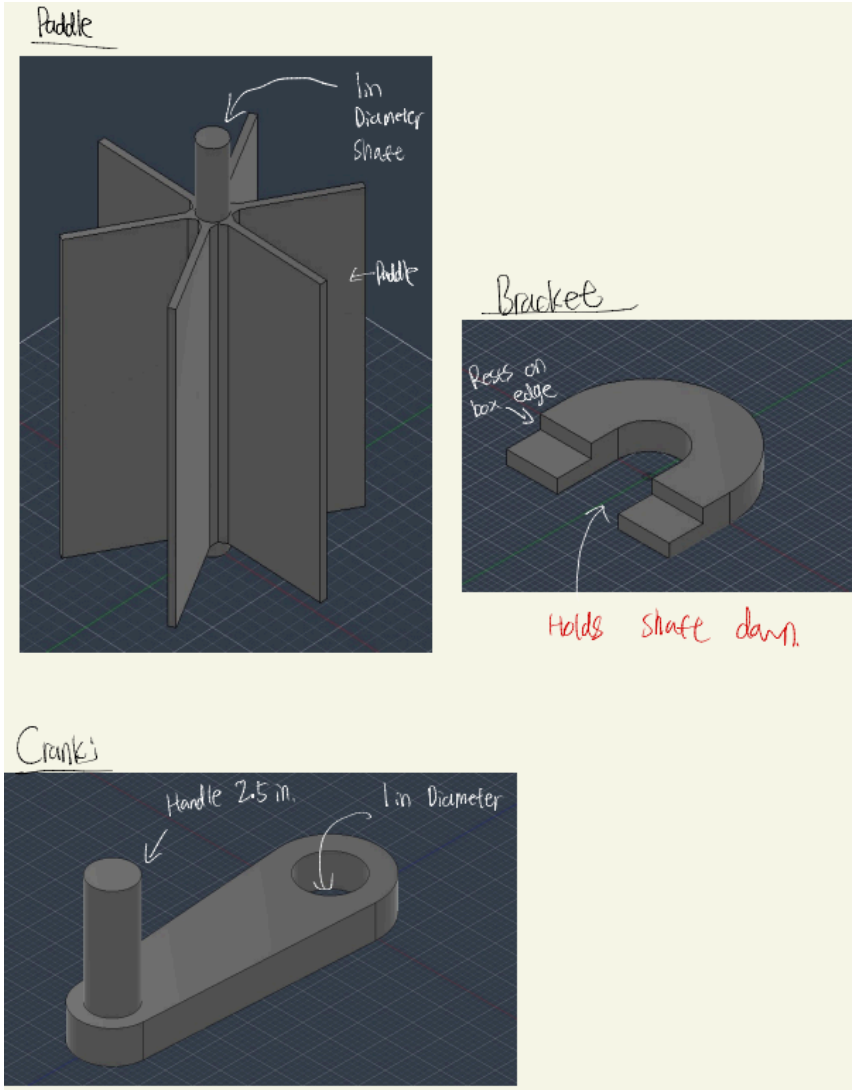


Assembly Instruction (CAD Parts did not arrive in time so these are instructions of our ODE 5 Lab prototype)

1. For the paddle we reused one of the cardboard tubes and cut wood panels to attach onto it using hot glue and string
2. The mounting brackets were also cut from scrap wood along with the crank.
3. To put everything together, first we line up the paddle at the right location on top of the storage container.
4. Put the mounting brackets in place and tape it to the side of the box to secure the bracket in place

5. Hot glue the crank onto the longer side of the shaft and rotating it will allow the paddle to the turn and sift SLFs





Parts List 3D prints -

Parts Name	Qty	Material	Dimensions
Paddle	1	PLA Plastic	Shaft ~12.5 in length, 1 in. diameter The Paddles are ~10in
Mounting Bracket	2	PLA Plastic	1 in. semicircle on the top to hold shaft in place
Crank	1	PLA Plastic	~ 1 in diameter to mount to shaft ~2.5 in. handle ~ 4 in length

Lab Makeshift Prototype Parts List -

Parts Name	**Qty**	**Material**	**Dimensions**
Paddle	1	Cardboard Shaft, Wood panels	Panels are ~3”x9” Shaft: ~13 in. length
Crank	1	Cardboard, wood	Handle: ~ 1.5 in Length ~ 4 in
Mounting Bracket	2	Wood	2 in. opening to accommodate the makeshift shaft

****Note****
Despite submitting our CAD files to RPL 4 days prior, the CAD components were still being processed in the RPL at the time prototypes were checked off in lab. Instead, we presented/designed a substitute component made of wood and cardboard of the same specifications, which were sufficient to complete our design tests. We will re-run tests to validate on our printed prototypes once they are manufactured.

Design tests:

1. Motion:
   Goal: Paddle wheel rotates smoothly and continuously.
   Test method: Rotate the crank 10 full revolutions at ~20 rpm (one revolution every ~3 s) while observing the paddle wheel.
   Measurements:
   a. Paddle completes 10/10 revolutions (1:1 with crank)
   b. No visible slip between crankshaft and paddle hub
   c. No stoppage or sticking during rotation
   Result: Paddle rotated 1:1 with the crank for 10 revolutions with no binding.
   Next prototype: We were planning to power the motion of the paddle with a motor, which will replace the crank.
2. Interference:
   Goal: No collisions between moving parts (paddles) and the tank, bracket.
   Test method: Rotate the assembly 10 full revolutions while checking all clearance zones (paddles to walls, shaft to bracket, paddles to waterline marker).
   Measurements:
   a. 0 contacts with any non-intended surfaces (tank walls, chute, bracket edges)
   b. Minimum clearance ≥ 5 mm between paddles and tank walls throughout rotation
   Result: No contact observed, although one had to be really careful when rotating the paddles since the brackets were made of wood and attached using tape.
   Next prototype: We will use the 3D printed parts instead of the wooden parts attached using tape for a more rigid and reliable design.

3. Fastening and Stability

Goal: Verify that the crank and the paddle are securely mounted to the plastic box so that it does not translate during rotation, and the box remains rigid under the applied crank load.

Test: We tested this by rotating the crank/paddle for 10 full rotations and observed the box, mount, crank/paddle for translational movement.

Pass criteria:

a. Box does not visibly flex or shift
b. Shaft stays seated (doesn't walk out)

Results:There was partial translational movement in both directions along the bracket.

Next prototype: Replace tape mounting with through-bolts + washers/clamps to reduce shaft movement; target bracket translation ≤ 2 mm and shaft wobble ≤ 5 mm over 10 revolutions.

4. Repeat use/cycling:

Goal: Prototype doesn't loosen or degrade over repeated use.

Test: Rotate at ~60 rpm for 1 minute (or as fast as you can consistently turn).

Pass criteria:

a. No cracks in paddles/hub/bracket
b. No tape peeling / bracket loosening
c. No increase in wobble by more than 2 mm compared to pre-test

Results: The prototype functioned as expected with no damage or deviation.

Next prototype: We will use the 3D-printed parts to improve stiffness and alignment, then repeat the cycling test with the box filled to the operating water level (same rpm/time) to verify the mount does not loosen, the shaft does not wobble more than 2 mm, and the paddle maintains clearance without splashing or contacting the walls.

Success Criteria:

Context:
Our project tackles the presence of SLFs in post-harvested grape/grape juice mixtures by 1) sorting SLF and solid grape by density once submerged into a sugar water solution, and 2) skimming the surface for SLF with a paddle and discarding them from the main compartment.

Criteria:

1. Brackets are securely mounted on the box to prevent any unwanted motion. The box can support the load from turning the crank and runs for 3 consecutive trials without stalling or requiring manual clearing
2. Paddles can effectively extract SLFs from the box (target SLF removal - 80%, target grape loss - 2%). This criterion is relevant to an exhibit day demonstration. It is incorporated in the final presentation, as our ultimate goal is to test the efficiency of our design, and testing this criterion will have a direct relation to the final presentation.
3. When operating at the target speed, the paddle wheel engages only the top surface layer and does not lift grape analogs. In a control test with grapes only, the system transfers 0 grapes to the discard compartment over 60 s of operation.